Carles Navarro

AI Scientist

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I am an AI scientist who works on machine learning for drug discovery. My research started with geometric deep learning, generative chemistry and reinforcement learning. Now I use this experience to build multi-agent LLM systems that plan, run and evaluate scientific experiments. In all my work, I take ideas from research to production with simple design and rigorous evaluation.

Experience

2025–Present **AI Scientist**, *Acellera Labs*, Barcelona
- Lead the development of PlayMolecule AI from research prototype to production, serving 2,000+ users and enterprise deployments in the pharmaceutical industry.
- Built a production multi-agent architecture for scientific discovery, with secure code execution and computational chemistry workloads on remote HPC clusters.
- Designed scientific evaluations and observability for AI agents, with containerized benchmarks from real research tasks, automated and expert grading, and iterative optimization.
- Shape the technical strategy and product roadmap with company leadership.

2022–2025 **Machine Learning Researcher, Industrial PhD**, *Acellera Labs*, Barcelona
- Designed 3D deep learning models for protein-ligand virtual screening and billion-scale molecular search.
- Trained chemical language models for molecular design, optimized with reinforcement learning.
- Applied reinforcement learning to train LLMs for reasoning on chemistry tasks.

2021–2022 **MSc Researcher, Computational Science Laboratory**, *Universitat Pompeu Fabra*, Barcelona
- Created graph neural network force fields for proteins, trained with differentiable molecular dynamics.

Education

2022–2026 **PhD in Biomedicine, Industrial Doctorate**, *Universitat Pompeu Fabra*, Barcelona
Thesis: *Agentic Intelligence for Molecular Modelling and Drug Discovery.* Supervisor: Prof. Gianni De Fabritiis.

2020–2022 **MSc in Bioinformatics for Health Sciences**, *Universitat Pompeu Fabra*, Barcelona

2015–2019 **BSc in Chemistry**, *Universitat Rovira i Virgili*, Tarragona

2019–Present **BSc in Physics, part time**, *UNED*, Spain

Skills

AI and ML: LLM agents, multi-agent systems, agent evaluation, reinforcement learning, generative models, geometric deep learning, contrastive learning

LLM stack: OpenAI, Anthropic and Gemini APIs, vLLM, Hugging Face Transformers, TRL, LoRA, MCP, Harbor

Programming: Python, TypeScript, SQL, C

Frameworks: PyTorch, PyTorch Geometric, TorchRL, FastAPI, React

Infrastructure: Linux, Docker, SLURM, GPU clusters, CI/CD

Languages: Catalan (native), Spanish (native), English (fluent)

Publications

2026 **C. Navarro**, M. Torrens-Fontanals, P. Thölke, S. Doerr, G. De Fabritiis. Speak to a Protein: An Interactive Multimodal Co-Scientist for Protein Analysis. *J. Chem. Inf. Model.*

2026 M. C. Thomas, R. Astudillo, Y. S. Raouf, **C. Navarro**, G. De Fabritiis, A. Bender. Generative Virtual Screening: From Search to Generate and Back. *Digital Discovery.*

2026 **C. Navarro**, P. Thölke, G. De Fabritiis. Structure-Guided Molecular Design with Contrastive 3D Protein-Ligand Learning. *arXiv:2604.19562.*

2024 N. Schapin, **C. Navarro**, A. Bou, G. De Fabritiis. On Machine Learning Approaches for Protein-Ligand Binding Affinity Prediction. *arXiv:2407.19073.*

2024 A. Bou, M. Thomas, S. Dittert, **C. Navarro**, et al., G. De Fabritiis. ACEGEN: Reinforcement Learning of Generative Chemical Agents for Drug Discovery. *J. Chem. Inf. Model.*

2023 **C. Navarro**, M. Majewski, G. De Fabritiis. Top-Down Machine Learning of Coarse-Grained Protein Force Fields. *J. Chem. Theory Comput.*